

PROJECT DOCUMENTATION IN POWER BI

RESTAURANT ORDER

Project Overview

In today's competitive food and restaurant industry, data analysis plays a major role in understanding customer preferences, improving sales performance, and making better business decisions. This project focuses on analyzing restaurant order data to uncover meaningful business insights related to customer preferences, sales performance, menu effectiveness, and operational trends. The analysis was performed using Microsoft Power BI to transform raw restaurant transaction data into interactive dashboards and visual reports.

The primary goal of this project is to help restaurant management make data-driven decisions by identifying top-performing menu items, high-revenue cuisine categories, customer ordering behavior, and low-performing products that require improvement strategies.

The dashboard provides a comprehensive view of restaurant operations through key performance indicators (KPIs), trend analysis, and category-level performance tracking. Various visualizations such as bar charts, pie charts, line charts, and KPI cards were used to analyze sales trends, order distribution, and menu popularity.

Objective

The main objective of this project is to analyze restaurant sales and customer ordering behavior using Power BI in order to generate meaningful business insights from transactional data.

1. To identify the most and least ordered menu items across the dataset.
2. Evaluate category-wise performance.
3. Monitor and optimize weekly Average Order Value trends.
4. Understand sales patterns across different time periods.
5. Identifying Risky Items and Best Performers by price and quantity.
6. Analyze spend tier contribution across time of day to optimize menu and operations.

The insights generated from this project can support strategic business decisions such as inventory planning, promotional campaigns, pricing strategies, menu redesign, and operational efficiency improvements.

Overall, this project demonstrates how business intelligence and data analytics can help restaurants improve customer satisfaction, increase profitability, and optimize overall business performance.

Data Sources

- **Dataset Source:** Mavenanalytics.io
- **Timeline:** 2023

- **Data Size:** 10,000+ rows and 7 columns
- **Domain:** Foods & Beverages

Data Cleaning (Preprocessing)

The following preprocessing steps were performed to ensure data quality:

- Handled missing values using Mean, Median, and Mode
- Removed duplicate records
- Corrected and changed data types
- Renamed columns for better readability

Tools Used:

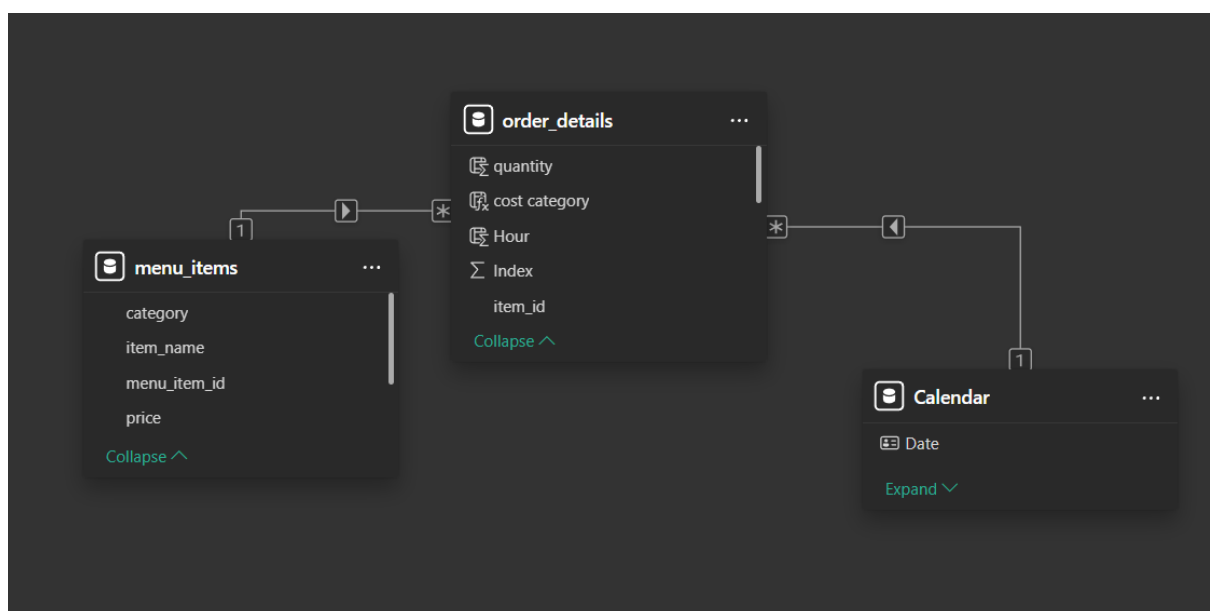
- Excel
- Power BI (Power Query)

Data Transformation

To enhance analysis, the following transformations were applied:

- Created new calculated columns
- Developed calculated tables
- Derived key performance indicators (KPIs) like- Total Revenue, Total Orders, Total Item Sold, Average Order Value.

Data Modelling



This is a **Power BI data model** with 3 tables and their relationships:

Tables:

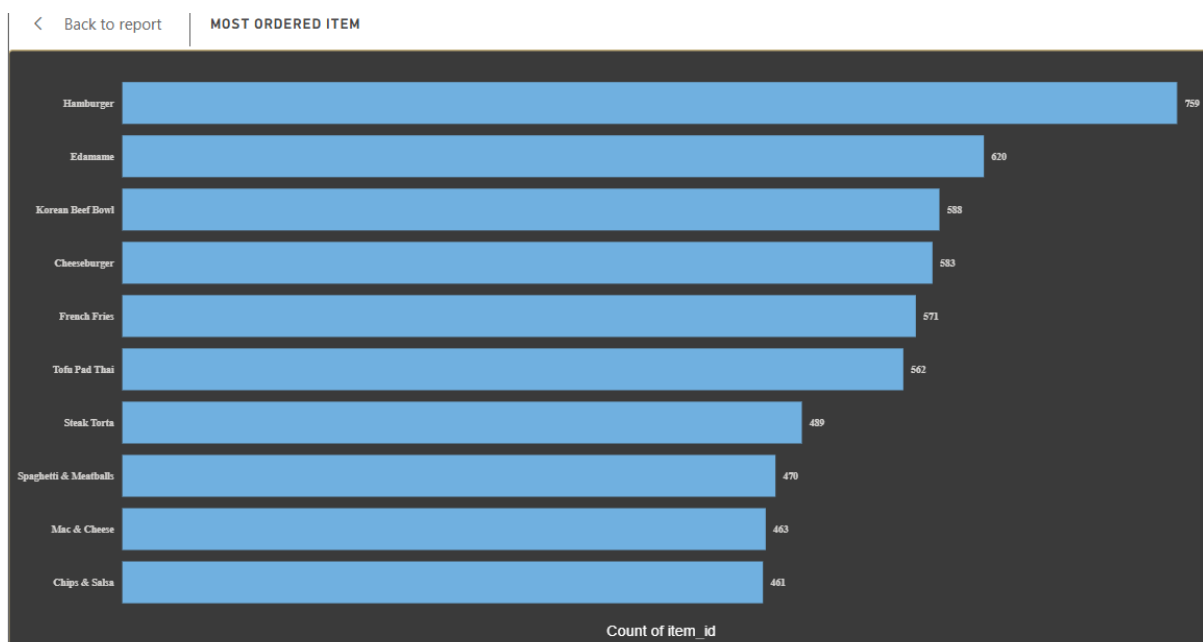
- **menu_items** - stores the menu catalog with category, item_name, menu_item_id, and price
- **order_details** - the fact/transaction table capturing quantity, cost category, Hour, Index, and item_id
- **Calendar** - a standard date dimension table with a Date column

Relationships:

- menu_items → order_details : **One-to-Many (1:*)** — one menu item can appear in many order lines, linked via item_id / menu_item_id
- Calendar → order_details : **One-to-Many (1:*)** — one date can have many orders, linked via the Date / Hour fields

Design Pattern: This follows a classic **Star Schema** — order_details sits at the center as the **fact table**, while menu_items and Calendar act as **dimension tables**, enabling time-based and category-based slicing of sales/order data.

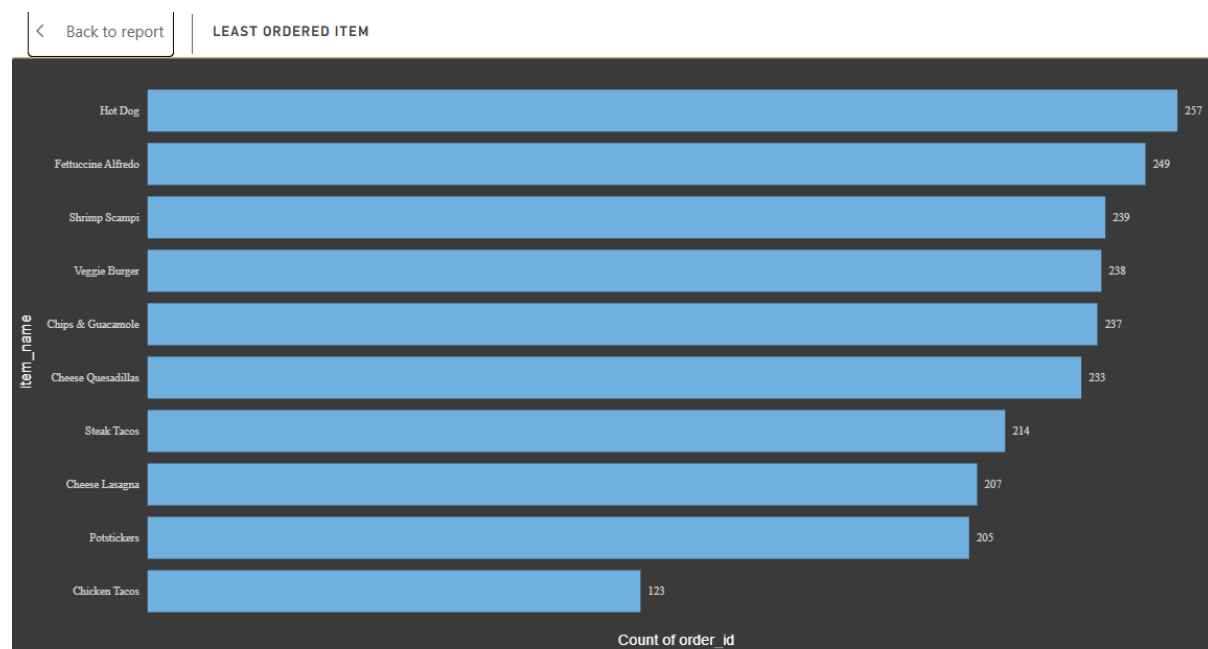
Key Insights From Most Ordered Item



- The most ordered items are **Hamburger (759 orders)**, **Edamame (620)**, and **Korean Beef Bowl (588)**. Classic fast-food items like burgers and fries dominate, while Asian-inspired dishes (Edamame, Pad Thai) also show strong demand.

- High performance is driven by **broad appeal, affordability, and quick preparation**. Items like Hamburger and Cheeseburger are staples, while Asian dishes may reflect growing customer interest in healthier or diverse options. The balance of comfort food and global cuisine explains their popularity.
- These items are likely to remain consistent best-sellers. Demand for Asian-inspired dishes may increase further, suggesting growth potential. Traditional fast-food items will continue to anchor sales but may plateau without innovation.
- Maintain strong supply and promotion of top sellers to avoid stockouts. Introduce **premium versions or combo deals** for high-demand items to lift Average Order Value. Use insights from Asian cuisine's popularity to expand menu variety and attract health-conscious customers.

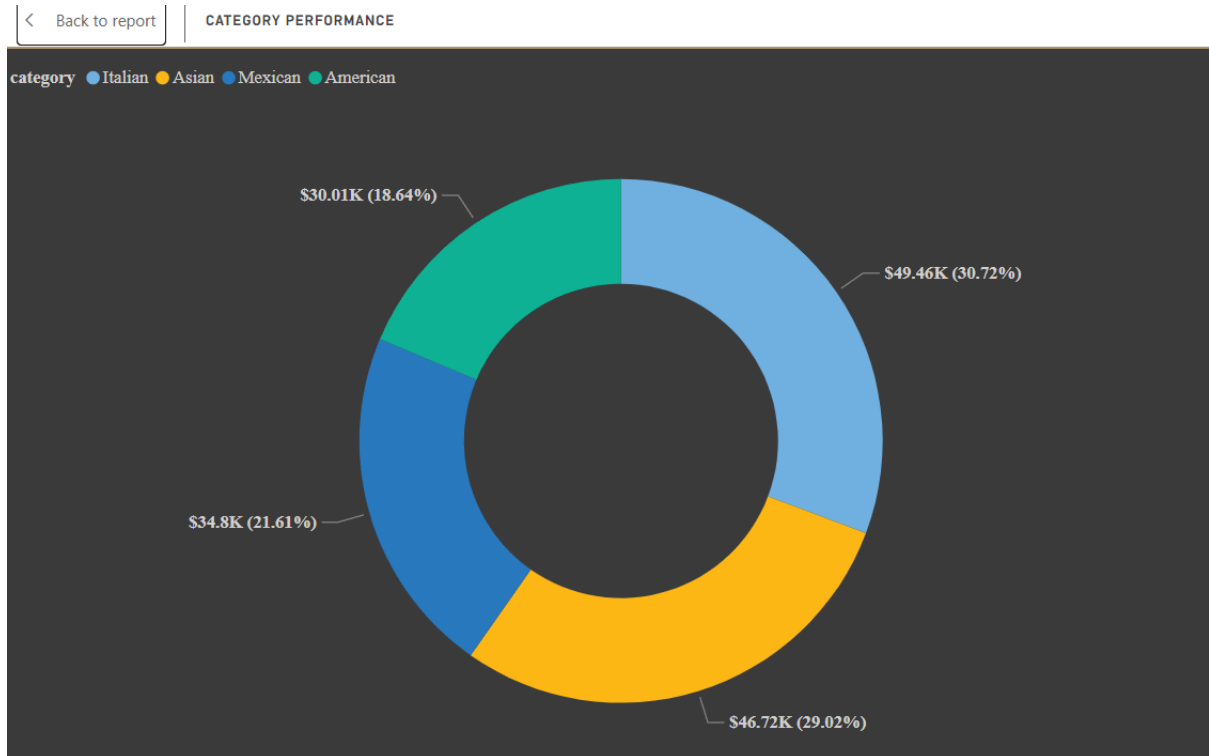
Key Insights From Least Ordered Item



- The least ordered items include **Hot Dog (257 orders)**, **Fettuccine Alfredo (249)**, and **Chicken Tacos (123)**. These products show significantly lower demand compared to top sellers, indicating weak customer preference.
- Low performance may be due to factors such as limited menu appeal, higher preparation time, pricing mismatch, or lack of visibility in promotions. For example, niche items like Shrimp Scampi or Cheese Lasagna may not align with mainstream customer tastes.

- If trends continue, these items will remain underperformers, potentially leading to wasted inventory or menu inefficiency. Without intervention, their contribution to revenue will stay minimal.
- Consider **menu redesign** (remove or replace persistently low sellers), **bundling strategies** (pair unpopular items with popular ones), or **targeted promotions** (discounts, seasonal offers) to test demand. Items with cultural or niche appeal could be repositioned as specialty offerings rather than regular menu staples.

Key Insights From Category Performance(Cuisine Revenue Analysis)

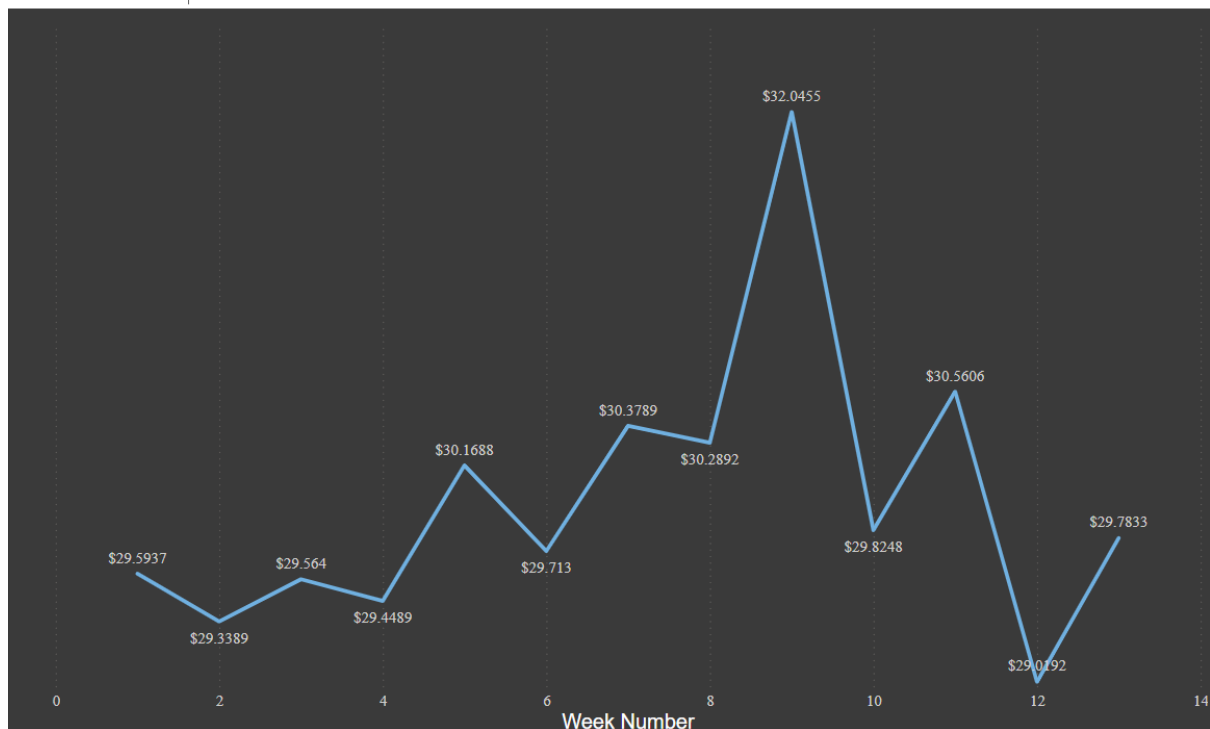


- Italian cuisine leads with **\$49,462.7**, followed closely by Asian (**\$46,720.65**). Mexican (**\$34,796.8**) and American (**\$30,011.9**) contribute less, showing clear variation in customer preferences.
- Italian and Asian cuisines likely perform better due to broader menu appeal, balanced pricing, and strong demand. Mexican and American may be underperforming due to limited variety, higher competition, or less frequent customer preference.
- If current trends continue, Italian and Asian will remain dominant contributors. Mexican cuisine shows potential for moderate growth, while American may stagnate unless menu innovation or promotions are introduced.
- Maintain strong focus on Italian and Asian items with upselling and premium bundles. Introduce targeted promotions for Mexican and American cuisines (combo offers, seasonal specials) to balance revenue contribution and diversify customer choices.

Key Insights from AOV Trend over Time

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AOV TREND OVER WEEKS

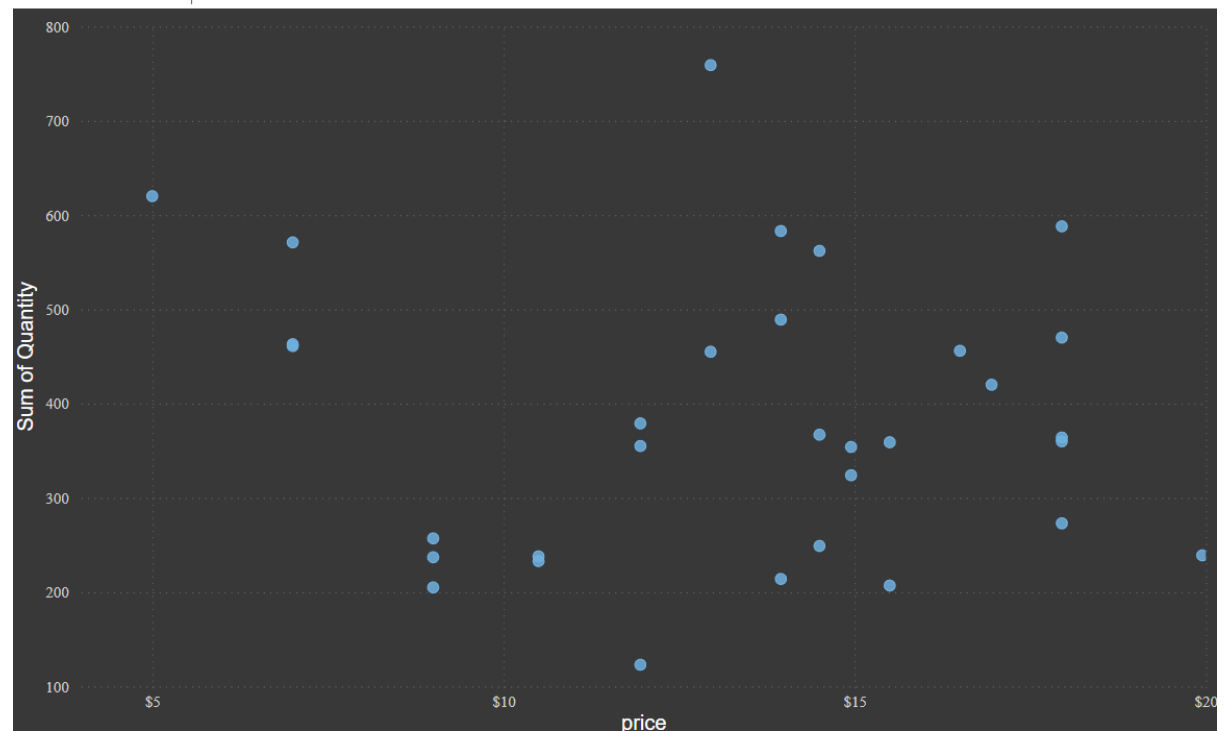


- The AOV fluctuates between **\$29.0** and **\$32.0** across 13 weeks. Week 9 shows the highest value (**\$32.05**), while Week 12 dips to the lowest (**\$29.01**). Overall, the average remains stable around **\$29.5–30.5**.
- The peaks (Weeks 5, 7, 9, 11) suggest promotional activity, menu mix changes, or customer preference shifts. The dip in Week 12 may indicate reduced demand, seasonal slowdown, or fewer high-value orders.
- If current patterns continue, AOV is likely to hover around **\$30** with occasional spikes above \$31 during promotional or festive weeks. Low-value weeks may reappear if menu variety or upselling is not maintained.
- Focus on replicating Week 9 strategies (bundling, upselling, premium items). Investigate Week 12's dip to prevent recurrence. Introduce targeted campaigns to push average order values consistently above \$31.

Key Insights from Risky Item and Best Performer

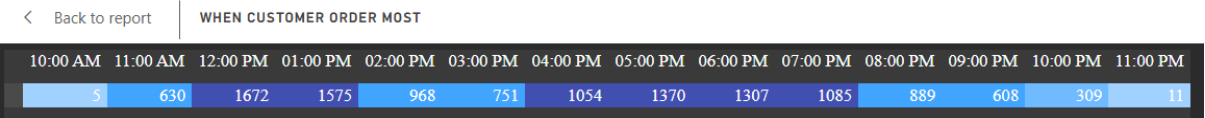
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RISKY ITEM AND BEST PERFORMER



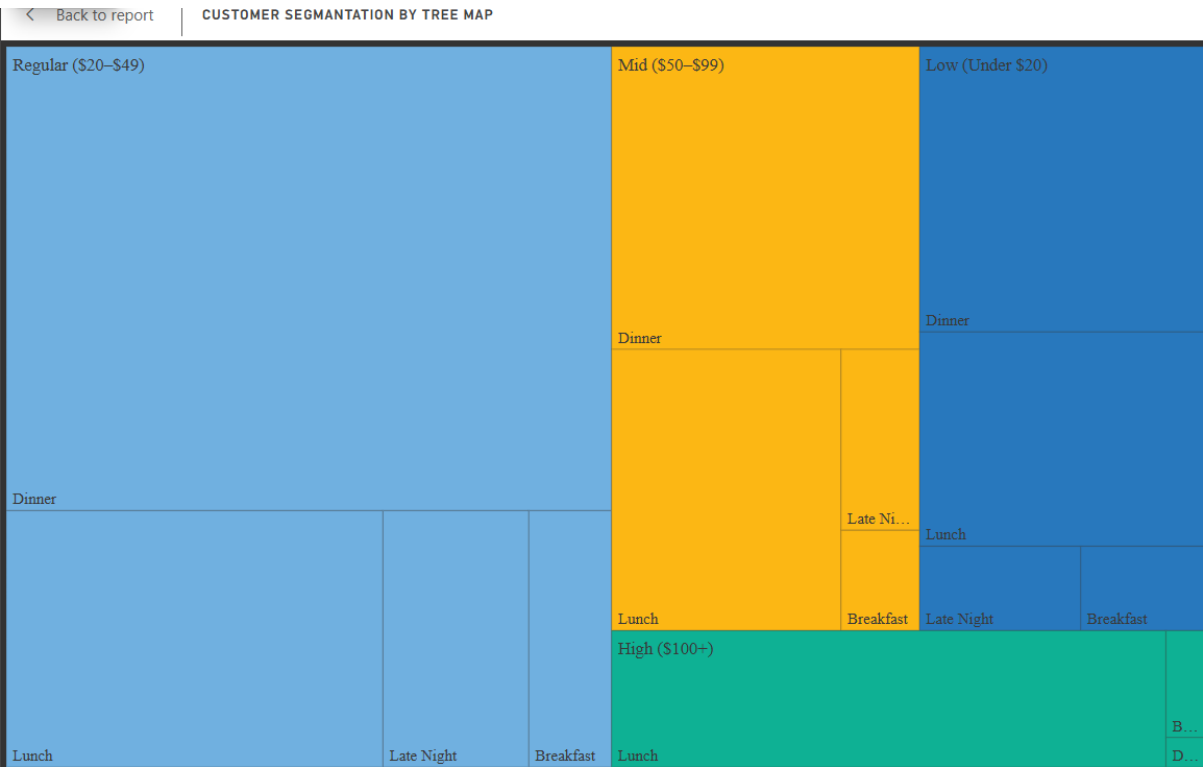
- The scatter plot shows items with varying demand and price points. High-quantity, low-price items like **Edamame (620 orders, \$5.00)** and **French Fries (571 orders, \$7.00)** dominate volume. Premium-priced items such as **Shrimp Scampi (\$19.95, 239 orders)** and **Spaghetti & Meatballs (\$17.95, 470 orders)** sell moderately but contribute higher per-order revenue.
- Low-priced staples succeed due to affordability and broad appeal. Mid-to-high priced items show mixed performance—some (Spaghetti & Meatballs, Korean Beef Bowl) balance strong demand with higher price, while others (Shrimp Scampi, Potstickers) struggle due to niche appeal or perceived value mismatch.
- If trends continue, **low-cost comfort foods** will remain consistent best-sellers. Mid-range items (\$12–15) like Cheeseburger and Mac & Cheese are likely to sustain demand. High-priced niche dishes may plateau unless repositioned or promoted.
- Maintain strong supply and marketing for high-volume, low-price items to secure steady traffic. For mid-to-high priced items, emphasize **value communication** (quality, portion size, premium experience). Consider **bundling strategies** (pairing niche items with popular ones) or **seasonal promotions** to lift demand for underperforming premium dishes.

Key Insights from When Customer Order Most



- The heatmap shows clear peaks at **12 PM (1,672 orders)** and **1 PM (1,575 orders)** for lunch, and again at **5 PM (1,370 orders)** and **6 PM (1,307 orders)** for dinner. Orders taper off after 9 PM, with minimal activity late night (11 PM: 11 orders).
- Lunch and dinner peaks align with typical meal times, confirming strong demand during these windows. The sharp decline after 9 PM suggests limited late-night menu appeal or reduced operating hours. The very low morning orders (10 AM: 5 orders) indicate customers rarely order before lunch.
- If trends continue, lunch and dinner will remain the busiest periods, consistently driving the majority of daily revenue. Late-night demand is unlikely to grow significantly without targeted promotions or menu adjustments.
- Optimize staffing and inventory around **12–1 PM** and **5–6 PM** to handle peak demand efficiently. Consider introducing **late-night offers or delivery-only specials** to capture untapped demand after 9 PM. Explore **breakfast or brunch menus** if expanding into earlier hours.

Key Insights from Customer Segmentation by Tree map



- Regular spenders (\$20–\$49) dominate revenue, especially at **Dinner (\$52,202)**. Mid-tier (\$50–\$99) contributes strongly at **Dinner (\$17,328.75)**, while High spenders

(\$100+) peak at **Lunch (\$14,053)**. Low spenders (<\$20) are consistent across all times, with notable contribution at **Dinner (\$15,330.3)**.

- Dinner attracts the highest spend across tiers, showing it as the prime revenue window. Lunch is strong for high-value customers, suggesting corporate or group dining. Breakfast and late-night show weaker contributions, likely due to limited menu appeal or customer habits.
- Dinner will continue to be the dominant revenue driver across tiers. High-value lunch orders are likely to remain stable, while late-night and breakfast segments may stay low unless targeted campaigns are introduced.
- Optimize staffing and inventory for **Dinner service**, ensuring premium and regular tier items are well-stocked. Enhance **Lunch offerings for high spenders** (premium combos, corporate packages). Introduce **breakfast bundles** or **late-night promotions** to lift underperforming segments.

Dashboard



CONCLUSION

The visual analyses across cuisines, product performance, pricing, time-based demand, and spend tiers collectively highlight clear patterns in customer behavior. Italian and Asian cuisines lead sales, top items like Hamburgers and Edamame drive demand, while low performers need repositioning. Lunch and dinner remain peak hours, and regular spenders dominate revenue with high-value customers contributing at lunch. Overall, these insights provide a strong foundation to prioritize menu expansion, balance pricing with demand, and optimize operations for sustainable growth.